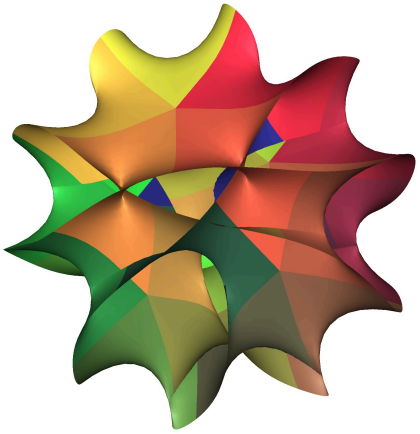
MATH742 - Geometric analysis



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1 Introduction

Theorem 1.1 $D \subseteq \mathbb{R}^n$ is open bounded with smooth boundary, $f \in C^\infty(\partial D)$, then Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } D \\ u = f & \text{on } \partial D \end{cases}$$

has a unique solution $u \in C^\infty(\overline{D})$. In the case $D = B(0,r)$, the solution is given by Poisson kernel

$$P[f](x) = \int_{\partial B(0,r)} f(\xi) \frac{r^2 - |x|^2}{r\omega_{n-1}|x-\xi|^2} d\sigma(\xi)$$

The uniqueness is guaranteed by integration by parts

Remark 1.2 Note that this always work as long as $\partial D \in C^\infty$

Theorem 1.3 M is a compact Riemannian manifold without boundary, $f \in C^\infty(M)$, if $\Delta u = f$ on M, then integration by parts demands $0 = \int_M \Delta u dx = \int_M f dx$, then $\Delta u = f$ on M has unique solution up to addting constants. Here $\Delta = \operatorname{Tr} \nabla^2_{X,Y}$ is the trace of Hessian, where $\nabla^2_{X,Y} = \nabla_X \nabla_Y - \nabla_Y X$

Theorem 1.4 M is a smooth manifold, the *de Rham complex* is

$$0 \to \mathcal{C}^\infty(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots$$

Define the cohomology to be *de Rham cohomology* $H^k_{\mathrm{dR}}(X, \mathbb{R})$, then

$$H^k_{\mathrm{dR}}(X, \mathbb{R}) \cong H^k_{\mathrm{sing}}(X, \mathbb{R}) = H^k(X, \mathbb{R})$$

Where $H^k(X, \mathbb{R})$ is the sheaf cohomology which is not so surprising by sheaf theory

Theorem 1.5 $s \in \Omega^k(X)$, the solution set S of $\Delta s = 0$ has $\dim S = \dim H^k(X, \mathbb{R})$. Actually $S \hookrightarrow H^k(X, \mathbb{R})$ with an explicit map

2 Laplace-Beltrami operator

Demailly VI.3: Hamonic forms and Hodge theory

(M, g) is a Riemannian manifold, $\dim M = m$, $(E, h) \to M$ is a $\mathbb{C}$ vector bundle, $\operatorname{rank} E = r$. $e_1, \cdots, e_r$ is an orthonormal basis of E, $e_1^*, \cdots, e_r^*$ is an orthonormal basis of E^*. $\xi_1, \cdots, \xi_m$ is an orthonormal basis of TM, $\xi_1^*, \cdots, \xi_m^*$ is an orthonormal basis of T^*M

Definition 2.1 (Hodge $*$ operator) The Riemannian volume form is $dV = \xi_1^* \wedge \cdots \wedge \xi_m^*$, $* : \bigwedge^p T^*M \to \bigwedge^{m-p} T^*M$ is defined such that $u \wedge * v = \langle u, v \rangle dV$

Theorem 2.2 $d : \mathcal{E}(M, \bigwedge^p T^*M) \to \mathcal{E}(M, \bigwedge^{p+1} T^*M)$ has a formal adjoint $d^* = (-1)^{mp+1} * d *$

Proof.

□

Definition 2.3 (Laplace-Beltrami operator) $\Delta = dd^* + d^*d$ which is self-adjoint and semi-positive definite

Example 2.4 (Laplace-Beltrami operator in Euclidean space)

References

Differential Analysis On Complex Manifolds - Raymond O. Wells

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